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BUSINESS INTELLIGENCE (BI)

For

Bachelor of Technology

In

Computer Science and Engineering

PROJECT REPORT :

GRADELA – GRADient Evaluation & Learned Academic Performance Radar

An Early-Warning Business Intelligence System for Predicting Student Academic Risk using Machine Learning in R

Continuous Assessment 3 (CA-3)

Mini Project Comprehensive Report

Submitted by

AISHI DE

PRN: 23070521008 | Section: A

PARTHIV ABHANI

PRN: 23070521106 | Section: B

Semester : VII

Year : 4

Academic Year : 2026-27

Subject Faculty

Dr. Snehlata Wankhade

Ph.D, Assistant Professor

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ABSTRACT

Higher education institutions generate a large amount of data about their students every semester including attendance registers, study logs, previous assessment marks, tutoring records and socio-economic background information. In most institutions, however, this data is analysed only after the final examinations have been graded. By the time a struggling student appears in a result sheet, the opportunity for meaningful academic support has already passed. Business Intelligence (BI) offers a systematic way of converting such operational data into timely, decision-ready information.

This project presents GRADELA: GRADient Evaluation & Learned Academic Performance Radar, an end-to-end early-warning Business Intelligence system developed in the R programming language. GRADELA uses the Student Performance Factors dataset containing 6,607 student records described by 20 academic, behavioural, socio-economic and environmental attributes. The raw data was cleaned through median and categorical-gap imputation, and a binary risk target was defined using the 25th percentile of the exam score distribution (Exam Score ≤ 65), which identified 2,131 students (32.25%) as High_Risk. Engineered features such as the Study Efficiency Index (SEI) were synthesised to capture the interaction between study volume, prior performance and sleep regularity.

Three supervised classifiers, i.e, Logistic Regression, Random Forest and Gradient Boosting (GBM) – were trained on a stratified 80/20 partition with 5-fold cross-validation, together with a CART decision tree for human-readable intervention rules and a Random Forest regressor for exact score forecasting. On the held-out test cohort of 1,321 students, the production Random Forest achieved 91.14% accuracy, 95.42% specificity and an ROC-AUC of 0.967. Because missing an at-risk student is far more costly than offering unnecessary tutoring, the decision boundary was recalibrated from $\tau = 0.50$ to a cost-sensitive $\tau^* = 0.38$, increasing recall on the at-risk class from 82.16% to 84.04% (F1 = 0.8658). The regressor explained 63.2% of exam-score variance with a mean absolute error of only 1.08 marks.

Feature-importance analysis shows that attendance and weekly study hours dominate the risk signal, while static demographic attributes contribute comparatively little, meaning that most risk is driven by modifiable behaviour. The model outputs are translated into a three-tier advising protocol (Critical, Moderate, Low), and an R dashboard with a hosted Vercel link.

Keywords: Business Intelligence, Early Warning System, Student Performance Prediction, Random Forest, Cost-Sensitive Classification, Threshold Calibration, Educational Data Mining, R Programming, Risk Stratification.



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1. INTRODUCTION

1.1 Background and Context

Student retention and timely academic support are among the most persistent challenges faced by higher education institutions. A considerable share of students experience academic difficulty well before their final examinations, yet traditional evaluation remains largely reactive: course marks, grades and pass/fail decisions are recorded only after teaching has ended. Remedial action that is triggered after a failed examination rarely changes the outcome of that semester.

At the same time, institutions already hold rich data on their students – attendance percentages, weekly study hours, previous assessment marks, participation in tutoring sessions, sleep habits, family background and access to learning resources. When these signals are analysed together, they can reveal the trajectory of a student long before a final grade is assigned. Educational Data Mining (EDM) and Machine Learning (ML) provide the analytical techniques required to do this at scale.

1.2 Motivation for the Study

Conventional early-warning mechanisms usually depend on a single rule, such as "attendance below 75%" or "failed the first quiz". Such univariate rules are easy to apply but ignore the interaction between behaviour, academic history and environment. A student may attend regularly but study very little; another may study long hours with highly irregular sleep. GRADELA was motivated by the need for a multivariate, data-driven radar that can flag the lower quartile of performers early enough for advisors to intervene, and that can explain why each student was flagged.

1.3 Problem Statement

The project addresses the following problem: **Given the behavioural, academic and contextual profile of a student that is available early in the semester, can we reliably predict whether the student will fall into the at-risk group (Exam Score ≤ 65), and can this prediction be converted into a prioritised, actionable intervention plan?**

Two practical constraints shape the solution. First, at-risk students form a minority (about one-third) of the cohort, so a model that simply optimises overall accuracy can still miss many vulnerable students. Second, the cost of errors is asymmetric – a missed at-risk student (False Negative) may fail or drop out, while a wrongly flagged student (False Positive) merely receives additional tutoring. The decision threshold therefore has to be tuned with these costs in mind.



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1.4 Business Intelligence Perspective

Although GRADELA uses machine learning, it is fundamentally a Business Intelligence solution. The "business" in this context is the academic institution, the key performance indicator is student success, and the decision-makers are faculty advisors and academic administrators. GRADELA follows the classic BI value chain shown in Fig. 1: raw operational data is transformed into information, information is converted into predictive knowledge, and knowledge is delivered as decisions in the form of tiered intervention rosters and interactive dashboards.

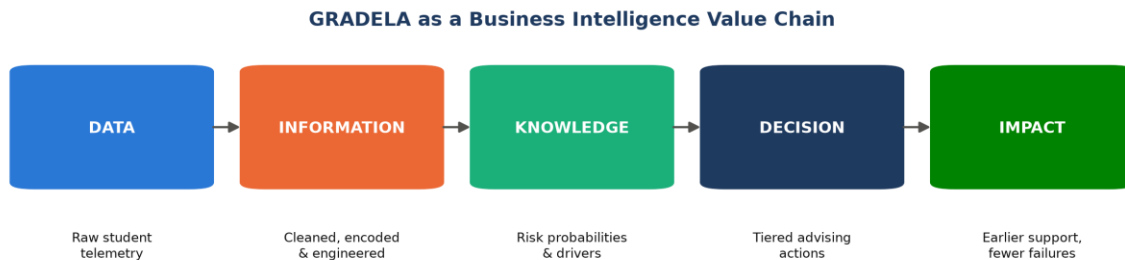


Fig. 1. GRADELA positioned as a Business Intelligence value chain



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2. PROJECT OBJECTIVES

The main objectives of the project are:

1. To ingest and audit a multidimensional student dataset of 6,607 records and 20 attributes using R.
2. To clean and prepare the data through column standardization, missing-value imputation and factor encoding.
3. To define an operational at-risk target using the 25th percentile of the exam score distribution (Exam Score ≤ 65).
4. To perform exploratory data analysis and identify the behavioural and contextual factors associated with academic risk.
5. To engineer composite behavioural features, including the Study Efficiency Index (SEI).
6. To train and benchmark Logistic Regression, Random Forest and Gradient Boosting classifiers using stratified cross-validation.
7. To calibrate a cost-sensitive decision threshold ($\tau^* = 0.38$) that maximises recall of at-risk students.
8. To forecast the exact exam score of each student using a Random Forest regressor.
9. To interpret the models through feature importance and a white-box decision tree that teachers can follow.
10. To stratify students into Critical, Moderate and Low risk tiers mapped to concrete advising actions.
11. To deliver the results through an exportable intervention roster, an R Shiny dashboard and a hosted web application.
12. To derive findings, problems and recommendations that can guide institutional academic-support strategy.



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3. LITERATURE REVIEW

Educational Data Mining has evolved from simple linear models towards ensemble and deep-learning approaches. Early studies established that attendance, previous grades and standardized test scores are strong baseline predictors of performance. As datasets grew to include behavioural and socio-demographic information, researchers shifted to non-parametric methods such as decision trees, support vector machines and Random Forests, which can capture non-linear relationships. Table 1 summarises representative work and its limitations relative to GRADELA.

Table 1. Review of related work

Reference	Method	Key Finding	Limitation
Cortez & Silva (2008)	Decision Trees, Random Forest, SVM	Secondary-school grades are strongly driven by earlier period grades (G1, G2).	Relies on late-stage marks; limited value for early intervention.
Romero & Ventura (2010)	Survey of EDM techniques	Classification, clustering and association mining are widely used to model learners.	Few studies address deployment or cost-sensitive decisions.
Marbouti et al. (2016)	Logistic Regression, Naïve Bayes, ensembles	At-risk engineering students can be flagged as early as the first weeks of a course.	Depends on structured weekly assessment data.
Breiman (2001)	Random Forests	Bagged, de-correlated trees reduce variance and provide feature importance.	Default threshold of 0.5 is not cost-aware.
Elkan (2001)	Cost-sensitive learning	Optimal decisions shift the probability threshold according to misclassification costs.	General theory; not specific to education.
GRADELA (2026)	LR, RF, GBM, CART in $R + \tau^*$ calibration	Behavioural factors dominate risk; $\tau^* = 0.38$ raises recall to 84.04%.	Single-institution style dataset; no longitudinal tracking yet.

The review highlights three gaps that GRADELA addresses: (i) most systems optimise symmetric accuracy and therefore under-serve the minority at-risk class, (ii) many models depend on late-semester assessments, and (iii) predictions are rarely converted into an operational intervention workflow that non-technical staff can use.



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4. DATASET DESCRIPTION

The project uses the **Student Performance Factors** dataset (Student_Performance_Factors_Dataset.csv). It contains 6,607 student records and 20 columns covering academic engagement, prior performance, family background, school environment and lifestyle. The final column, Exam_Score, is the outcome variable from which the risk target is derived.

Table 2. Dataset dimensions

Category	Details
Student records	6,607
Attributes	20 (19 predictors + Exam_Score)
Numerical attributes	7 – Hours_Studied, Attendance, Sleep_Hours, Previous_Scores, Tutoring_Sessions, Physical_Activity, Exam_Score
Categorical attributes	13 – involvement, resources, motivation, income, teacher quality, school type, peer influence, etc.
Target (regression)	Exam_Score (mean 67.24, SD 3.89, range 55 – 101)
Target (classification)	risk_status: High_Risk (≤ 65) vs Normal (> 65)
Missing values	235 cells across 3 categorical columns

Table 3 lists every field in the dataset with its type and meaning. The attributes can be grouped into four families: **academic engagement** (hours studied, attendance, tutoring), **academic history** (previous scores), **home and socio-economic context** (parental involvement, family income, parental education, internet access, distance from home) and **personal / lifestyle factors** (sleep, physical activity, motivation, peer influence, learning disabilities).



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Table 3. Dataset field description

Field	Type	Description
Hours_Studied	Numeric	Hours spent studying per week (1 – 44)
Attendance	Numeric	Percentage of classes attended (60 – 100)
Parental_Involvement	Ordinal	Low / Medium / High
Access_to_Resources	Ordinal	Availability of educational resources
Extracurricular_Activities	Binary	Participation in extracurricular activities
Sleep_Hours	Numeric	Average nightly sleep (4 – 10 hours)
Previous_Scores	Numeric	Score in previous examinations (50 – 100)
Motivation_Level	Ordinal	Low / Medium / High
Internet_Access	Binary	Yes / No
Tutoring_Sessions	Numeric	Tutoring sessions attended per month (0 – 8)
Family_Income	Ordinal	Low / Medium / High
Teacher_Quality	Ordinal	Low / Medium / High (78 missing)
School_Type	Nominal	Public / Private
Peer_Influence	Nominal	Negative / Neutral / Positive
Physical_Activity	Numeric	Hours of physical activity per week (0 – 6)
Learning_Disabilities	Binary	Yes / No
Parental_Education_Level	Ordinal	High School / College / Postgraduate (90 missing)
Distance_from_Home	Ordinal	Near / Moderate / Far (67 missing)
Gender	Nominal	Male / Female
Exam_Score	Numeric	Final exam score – outcome variable

Table 4. Descriptive statistics of numerical attributes

Attribute	Mean	Std. Dev.	Min	Median	Max
Hours_Studied	19.98	5.99	1	20	44
Attendance	79.98	11.55	60	80	100
Sleep_Hours	7.03	1.47	4	7	10
Previous_Scores	75.07	14.40	50	75	100
Tutoring_Sessions	1.49	1.23	0	1	8
Physical_Activity	2.97	1.03	0	3	6
Exam_Score	67.24	3.89	55	67	101



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5. DATA PREPARATION

Data preparation directly determines the reliability of every downstream model and dashboard. The preparation stage was implemented in **R/01_data_preprocessing.R** using the tidyverse and caret packages. All parameters (random seed, split ratio, risk quantile) are read from **config.yml**, which makes the pipeline fully reproducible.

5.1 Column Standardization

All column names were converted to lower-case snake_case (for example, Hours_Studied → hours_studied) using `tolower(gsub("[ .]", "_", names(df)))`. Consistent naming prevents errors when the same columns are referenced in the feature-engineering, training, inference and dashboard scripts, and allows new cohort files with slightly different formatting to be ingested by the batch inference engine.

5.2 Missing Value Treatment

A missingness audit identified 235 empty cells, all in categorical attributes. Because each affected column has less than 1.5% missing data, deleting rows would have discarded legitimate student records unnecessarily. Instead, a type-aware imputation strategy was used:

Table 5. Missing value audit and imputation scheme

Attribute	Missing Count	Missing %	Treatment
Teacher_Quality	78	1.18%	Explicit "Unknown" factor level
Parental_Education_Level	90	1.36%	Explicit "Unknown" factor level
Distance_from_Home	67	1.01%	Explicit "Unknown" factor level
All numerical attributes	0	0.00%	Median imputation rule applied defensively

Numerical gaps (none in the current file, but possible in future cohorts) are filled with the column median, which is robust to outliers such as the exam scores above 100. Categorical gaps are assigned an explicit "Unknown" level rather than the mode, so the model can learn whether missing information itself carries a signal, and so that no artificial inflation of the most common category is introduced. All character columns were then converted to R factors.

5.3 Duplicate and Consistency Checks

The dataset was checked for duplicate rows and none were found. Value ranges were validated (attendance 60–100%, sleep 4–10 hours, previous scores 50–100). A small number of exam scores above 100 were retained because they represent a legitimate long tail and do not affect the



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risk boundary at 65. Categorical text values were trimmed of whitespace to ensure that the same category is never split into two levels.

5.4 Risk Target Formulation

The continuous Exam_Score was converted into a binary early-warning target using the 25th percentile of the score distribution, configured as `risk_quantile_cutoff = 0.25` in `config.yml`. The resulting cut-off is **65.00**:

risk_status = High_Risk if Exam_Score $\leq$ 65, otherwise Normal

Because many students score exactly 65, the $\leq$ rule places 2,131 students (32.25%) in the High_Risk class and 4,476 (67.75%) in the Normal class, as shown in Fig. 2. This moderate imbalance is the reason why recall, rather than raw accuracy, is treated as the primary operational metric.

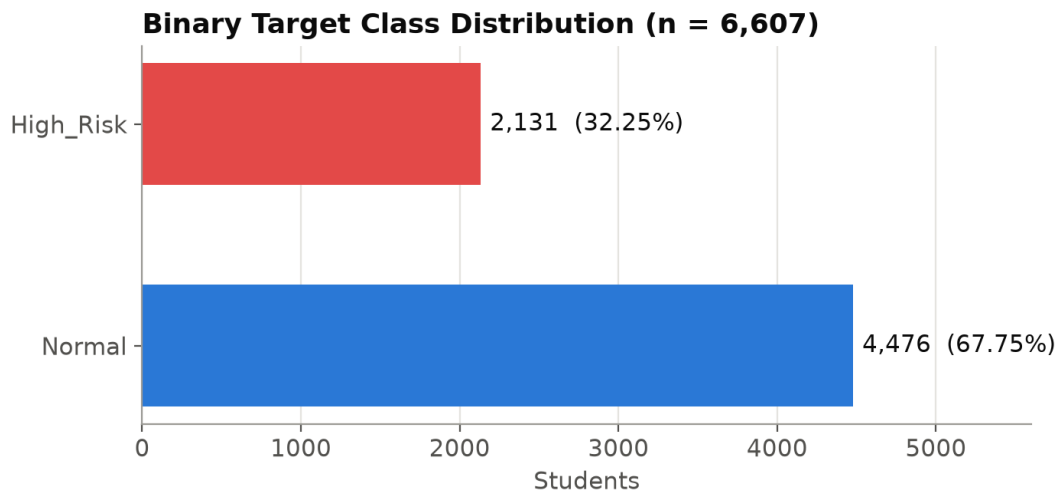


Fig. 2. Binary target class distribution after risk formulation

5.5 Stratified Partitioning

The cohort was split 80/20 into training and test partitions using `caret::createDataPartition` with seed 42. Stratification preserves the 32.25% High_Risk prevalence in both partitions, producing 5,286 training records and 1,321 test records. All metrics reported in this report are computed on the untouched test partition.

5.6 Overall Data Preparation Outcome

The preparation stage ensured that:

- Column names are standardized and consistent across all scripts.



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- All 6,607 records are retained – no student was deleted because of missing data.
- 235 categorical gaps are handled explicitly and transparently.
- No duplicate records exist in the dataset.
- A binary risk target with a data-driven cut-off of 65 is available.
- Stratified training and test partitions are exported to data/processed/ for reproducibility.

Key Outcomes of the Data Preparation Stage

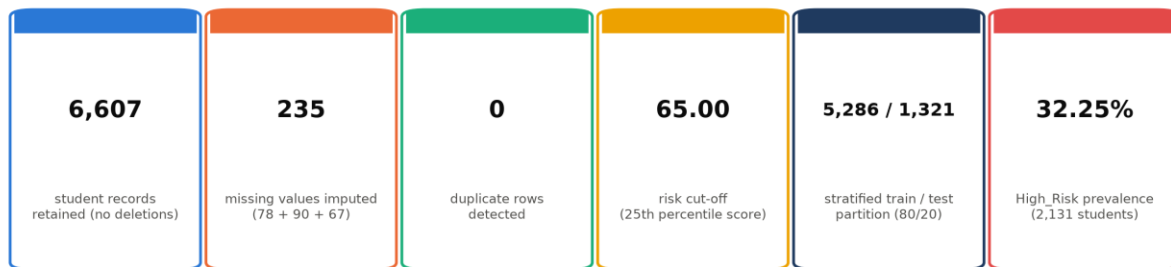


Fig. 3. Key outcomes of the data preparation stage



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6. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was carried out to understand the shape of the outcome variable, the relationships between numerical predictors, and how the at-risk rate varies across behavioural and contextual groups. These insights guided both the feature engineering and the interpretation of the final models.

6.1 Exam Score Distribution

Exam scores are concentrated in a narrow band between 60 and 75 with a mean of 67.24 and a standard deviation of only 3.89. Fig. 4 shows that the at-risk boundary at 65 cuts through the dense left shoulder of the distribution. Because the range is so narrow, a difference of only two or three marks separates an at-risk student from an average one – which explains why small behavioural changes can move a student across the boundary.

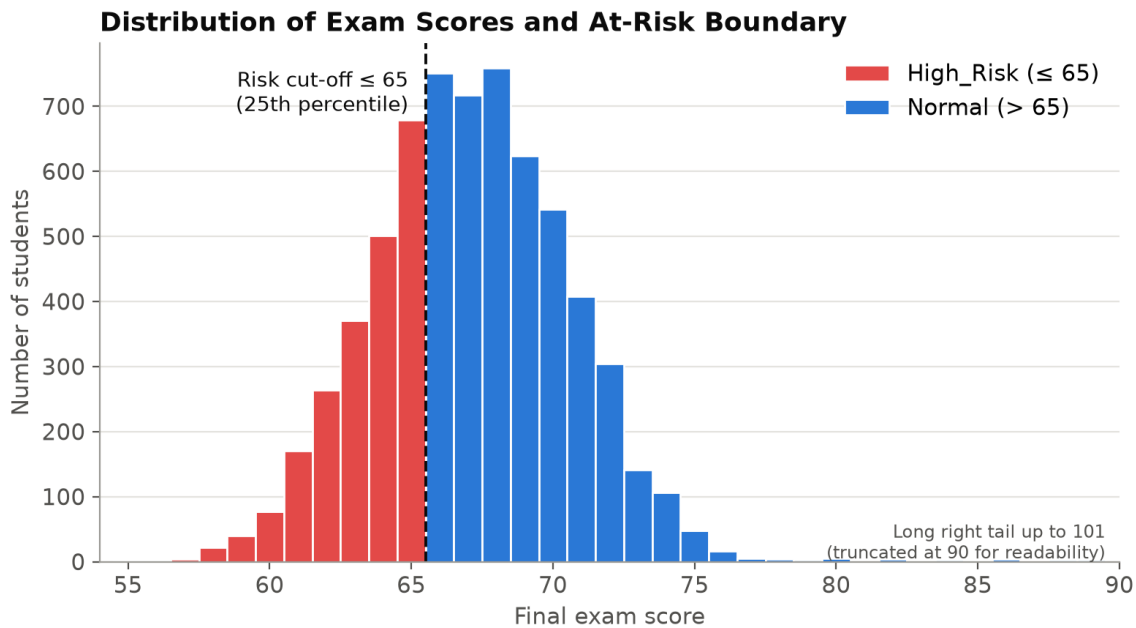


Fig. 4. Distribution of exam scores and the at-risk boundary

6.2 Correlation Analysis

Fig. 5 shows the correlation matrix of the numerical attributes. **Attendance** ($r = 0.58$) and **Hours_Studied** ($r = 0.45$) have the strongest positive correlation with Exam_Score, followed by Previous_Scores (0.18) and Tutoring_Sessions (0.16). Sleep_Hours and Physical_Activity show near-zero linear correlation. Importantly, predictors are almost uncorrelated with one another ($|r| \leq 0.02$), so multicollinearity is not a concern for either the linear or the tree-based models.



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GRADELA: Feature Correlation Matrix

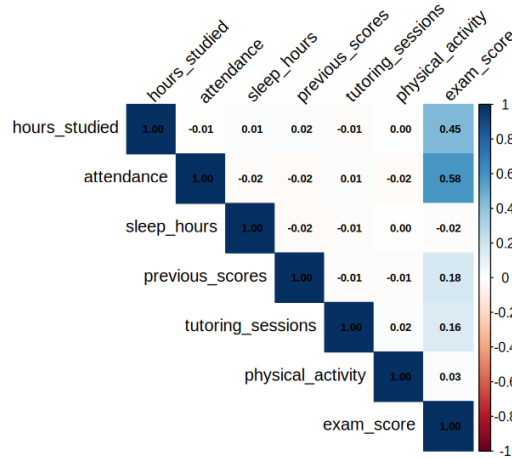


Fig. 5. Feature correlation matrix (corrplot, R)

6.3 Behavioural Indicators

Fig. 6 compares the three most important numerical indicators between the two classes. At-risk students attend on average **70.8%** of classes compared with **84.4%** for Normal students – a gap of almost 14 percentage points. They also study about **4.8 fewer hours per week** (16.7 vs 21.5) and have lower previous scores (71.9 vs 76.6). The non-overlapping interquartile ranges for attendance make it the clearest single warning signal in the dataset.

Behavioural Indicators: At-Risk vs. Normal Students

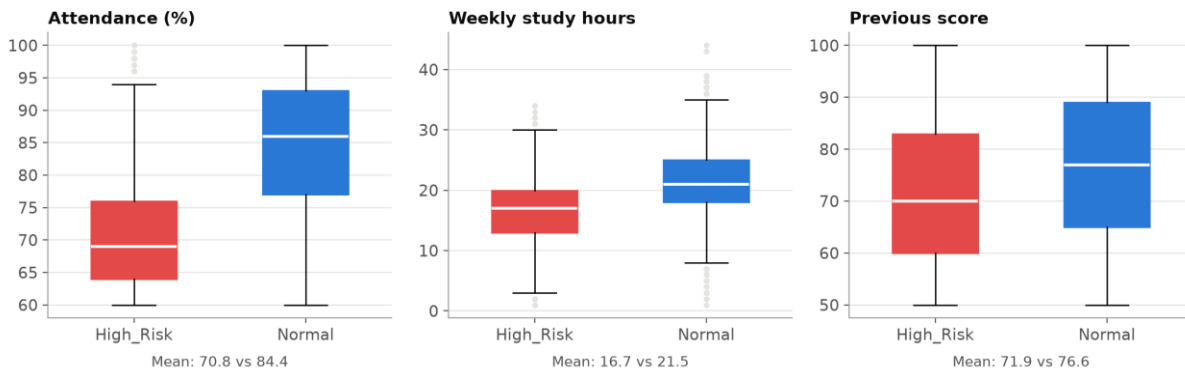


Fig. 6. Behavioural indicators of at-risk versus normal students



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6.4 Contextual and Socio-Economic Factors

Fig. 7 shows the percentage of at-risk students within each level of eight contextual factors. Red bars indicate levels with a risk rate clearly above the cohort average of 32.3%; blue bars indicate clearly protective levels.

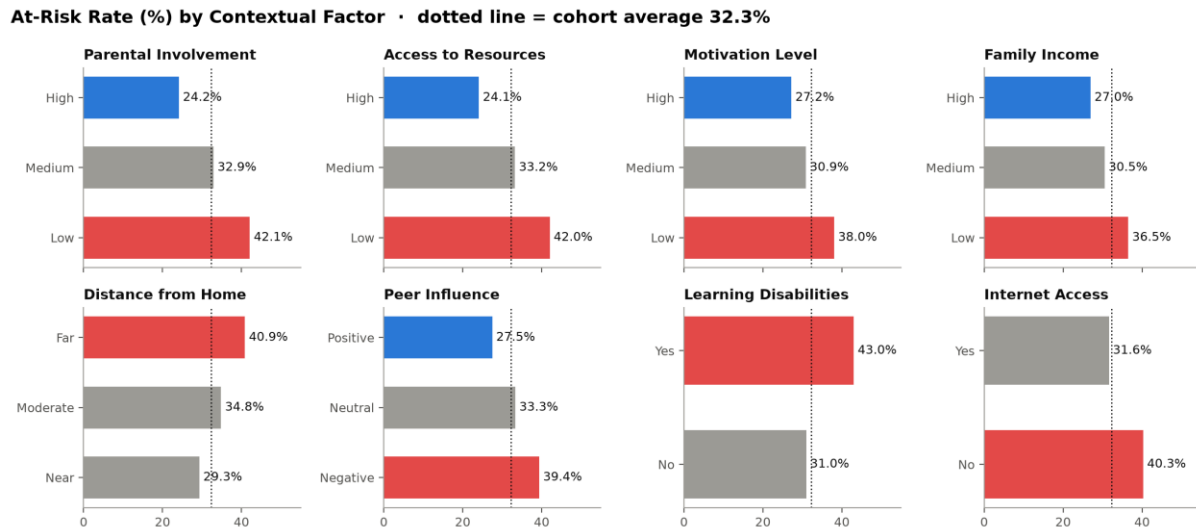


Fig. 7. At-risk rate by contextual and socio-economic factor

The analysis shows that:

- Students with **low parental involvement** (42.1%) or **low access to resources** (42.0%) are about 1.7 times as likely to be at risk as those with high involvement or access ($\approx 24\%$).
- Students with **learning disabilities** have an at-risk rate of 43.0% compared with 31.0% for other students.
- **Negative peer influence** (39.4%) and living **far from school** (40.9%) are associated with elevated risk.
- Lack of **internet access** raises the at-risk rate to 40.3%.
- Gender (32.6% vs 32.0%) and school type (31.4% vs 32.6%) show virtually no difference, confirming that risk is not driven by fixed demographic labels.



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7. FEATURE ENGINEERING

Raw attributes describe individual behaviours in isolation. In reality, the effect of studying depends on how well a student is rested and on their existing level of understanding. The script **R/02_feature_engineering.R** therefore synthesises four composite indicators.

7.1 Study Efficiency Index (SEI)

The Study Efficiency Index models how effectively study time is converted into learning, after adjusting for sleep irregularity:

$$SEI = (\text{Hours_Studied} \times \text{Previous_Scores}) / (\text{Sleep_Variance} + \epsilon), \quad \epsilon = 10^{-4}$$

where $\text{Sleep_Variance} = |\text{Sleep_Hours} - 7| + 0.8$ measures deviation from a regular seven-hour circadian baseline. A student who studies many hours, has a solid academic base and sleeps regularly receives a high SEI, whereas the same study volume combined with irregular sleep produces a lower value. The construction is illustrated in Fig. 8.

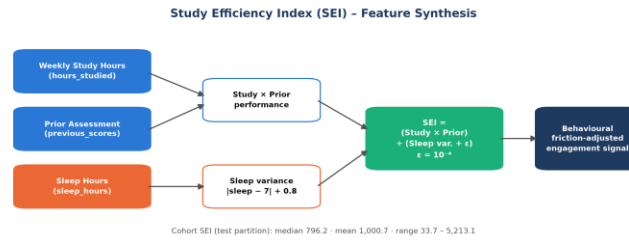


Fig. 8. Construction of the Study Efficiency Index (SEI)

7.2 Additional Engineered Features

Table 6. Engineered behavioural features

Feature	Formula	Purpose
sleep_variance	$ \text{sleep_hours} - 7 + 0.8$	Circadian irregularity relative to a 7-hour baseline
study_efficiency_index	$(\text{hours_studied} \times \text{previous_scores}) / (\text{sleep_variance} + \epsilon)$	Friction-adjusted study effectiveness
attendance_tutoring_synergy	$\text{attendance} \times (\text{tutoring_sessions} + 1)$	Combined effect of class presence and extra support
academic_engagement_ratio	$(\text{hours_studied} + 2 \times \text{tutoring_sessions}) / (\text{sleep_hours} + 1)$	Overall engagement relative to rest

These features are exported to `data/processed/gradela_features_engineered.csv` and used in the risk-alert dispatch stage to explain the primary driver behind each student's risk score.



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8. SYSTEM ARCHITECTURE AND METHODOLOGY

8.1 System Architecture

GRADELA is organised into five layers (Fig. 9). The **data layer** holds the raw cohort file, configuration and new batch files. The **processing layer** performs preprocessing, feature synthesis and partitioning in R. The **analytics layer** trains the model suite. The **decision layer** applies the calibrated threshold, forecasts scores and assigns risk tiers. Finally, the **presentation layer** delivers results to advisors through the intervention roster, an R Shiny dashboard and the hosted GRADELA AI web application.

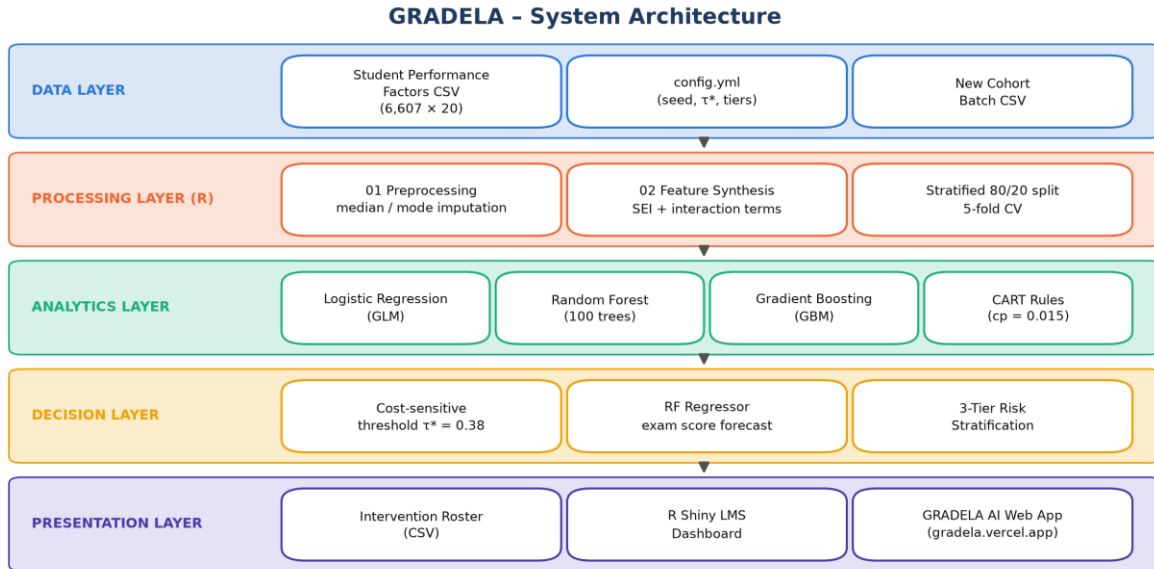


Fig. 9. Layered system architecture of GRADELA

8.2 Predictive Pipeline

The complete workflow is orchestrated by **run_pipeline.R**, which sources six numbered scripts in order. Each stage writes its outputs to disk so that it can be re-run or inspected independently.



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GRADELA End-to-End Predictive Pipeline (run_pipeline.R)

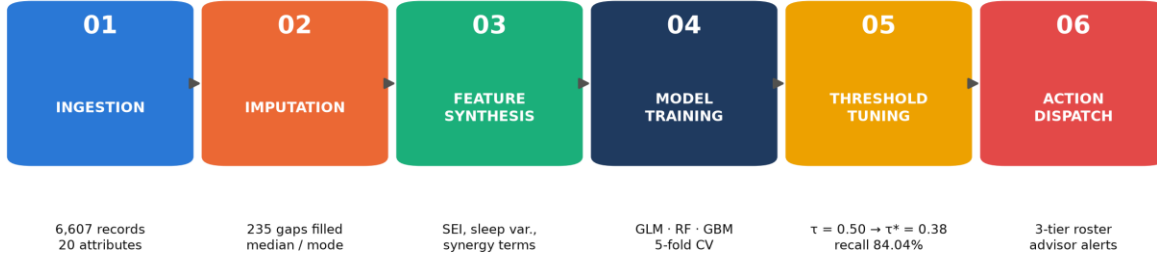


Fig. 10. End-to-end predictive pipeline orchestrated by run_pipeline.R

Table 7. Pipeline stages, scripts and outputs

Stage	Script	Output
01 Ingestion & preprocessing	R/01_data_preprocessing.R	Cleaned cohort, train/test partitions
02 Feature synthesis	R/02_feature_engineering.R	SEI and interaction features
03 Model training	R/03_model_training.R	GLM, RF, GBM, CART, RF regressor (.rds)
04 Threshold tuning	R/04_threshold_tuning.R	Sensitivity across τ , ROC-AUC
05 Alert dispatch	R/05_alert_dispatch.R	student_risk_alerts.csv, intervention roster
06 Dashboard	R/shiny_app.R	Interactive LMS early-warning dashboard

8.3 Model Suite

All classifiers were trained with `caret::train` using 5-fold cross-validation, class probabilities and the ROC metric (`twoClassSummary`) as the optimisation target.

1) Logistic Regression (GLM)

A generalized linear model with a binomial link serves as the interpretable baseline. It estimates the log-odds of being High_Risk as a linear combination of the predictors: $\log(p / (1 - p)) = \beta_0 + \sum \beta_j x_j$.

2) Random Forest Classifier

An ensemble of 100 de-correlated decision trees, each grown on a bootstrap sample with a random subset of features at each split. The forest averages the votes of all trees, which reduces variance, handles mixed numerical and categorical inputs natively and provides Gini-based feature importance. The Random Forest is GRADELA's production classifier.



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3) Gradient Boosting Machine (GBM)

A boosting ensemble that adds shallow trees sequentially, each one fitted to the pseudo-residuals of the previous ensemble (100 trees, interaction depth 3, shrinkage 0.1).

4) CART Decision Tree

A single rpart classification tree with complexity parameter $cp = 0.015$. It is not used for scoring but for generating transparent if-then rules that teachers can apply without running the model.

5) Random Forest Regressor

A 120-tree random forest that predicts the exact exam score, complementing the binary risk flag with a fine-grained forecast.

Table 8. Model suite configuration (config.yml)

Model	Role	Key Configuration
Logistic Regression	Linear baseline	family = binomial
Random Forest	Production classifier	ntree = 100, importance = TRUE
Gradient Boosting	Boosting benchmark	n.trees = 100, depth = 3, shrinkage = 0.1
CART	White-box rules	method = "class", cp = 0.015
RF Regressor	Score forecaster	ntree = 120

8.4 Algorithmic Pseudocode

```

Input : D = {(xi, yi)}, N = 6,607 students, P = 19 predictors
Output: Calibrated model M*, threshold tau*, risk tier for every student

1: D <- standardize_names(D); impute(numeric = median, categorical = "Unknown")
2: cutoff <- quantile(Exam_Score, 0.25) # = 65
3: Y_i <- High_Risk if Exam_Score_i <= cutoff else Normal
4: (D_train, D_test) <- stratified_split(D, 0.80, seed = 42)
5: for m in {GLM, RandomForest, GBM}:
6:   fit m on D_train with 5-fold CV, metric = ROC
7: M* <- RandomForest; p_hat <- M*(D_test)
8: tau* <- argmin_tau [ C_FN * (1 - Recall(tau)) + C_FP * (1 - Specificity(tau)) ] #
0.38
9: for each student j:
10:   if p_hat_j >= 0.60 -> Tier 1 (Critical)
11:   else if p_hat_j >= tau* -> Tier 2 (Moderate)
12:   else -> Tier 3 (Low)
13:   driver_j <- dominant_deficit (attendance, study volume, sleep, motivation,
tutoring)
14: export roster, alerts and dashboard data

```



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9. MODEL EVALUATION AND RESULTS

All results in this chapter are computed on the stratified test partition of **1,321 students** that was never used during training. The positive class is High_Risk.

9.1 Confusion Matrix Analysis

Fig. 11 shows the confusion matrix of the Random Forest at the default threshold of 0.50. Of the 426 truly at-risk students, 350 were correctly identified and 76 were missed. Only 41 of the 895 Normal students were incorrectly flagged.

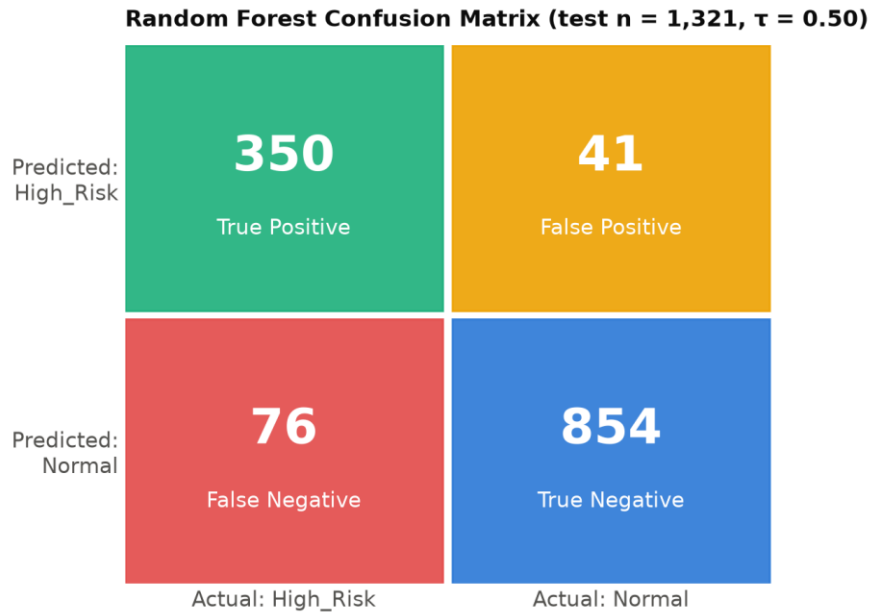


Fig. 11. Random Forest confusion matrix on the test cohort

Table 9. Random Forest classification metrics ($\tau = 0.50$)

Metric	Value	Interpretation
Accuracy	91.14%	95% CI: 89.48% – 92.62%
Sensitivity (Recall)	82.16%	Share of at-risk students identified
Specificity	95.42%	Share of normal students correctly cleared
Precision (PPV)	89.51%	Flagged students who are truly at risk
Negative Predictive Value	91.83%	Cleared students who are truly normal
Balanced Accuracy	88.79%	Mean of sensitivity and specificity
Cohen's Kappa	0.7929	Substantial agreement beyond chance
No-Information Rate	67.75%	Accuracy is significantly higher ($p < 2.2 \times 10^{-16}$)



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9.2 Comparative Benchmark

Table 10. Performance benchmark on the test partition (n = 1,321)

Algorithm	Accuracy	Sensitivity	Specificity	Precision	F1-Score	ROC-AUC
Random Forest	0.9114	0.8216	0.9542	0.8951	0.8568	0.9673
Gradient Boosting (GBM)	0.9031	0.7864	0.9587	0.9005	0.8396	0.9675
Logistic Regression	0.9765	0.9531	0.9877	0.9736	0.9632	0.9968
Random Forest @ $\tau^* = 0.38$	—	0.8404	—	0.8930	0.8658	0.9673

Logistic Regression achieved the highest scores on every metric. This is an important finding in itself: in this dataset the exam score is close to an additive combination of the predictors, so a linear model separates the classes very well. The two tree ensembles performed similarly to each other, with Random Forest offering the higher sensitivity of the two. Random Forest was retained as the production engine because it handles future non-linear cohorts without manual feature transformation, provides stable feature-importance rankings, shares its structure with the score regressor and supports the calibrated tier system; Chapter 14 discusses this trade-off further.

Fig. 1. Comparative Validation Performance Across Machine Learning Models

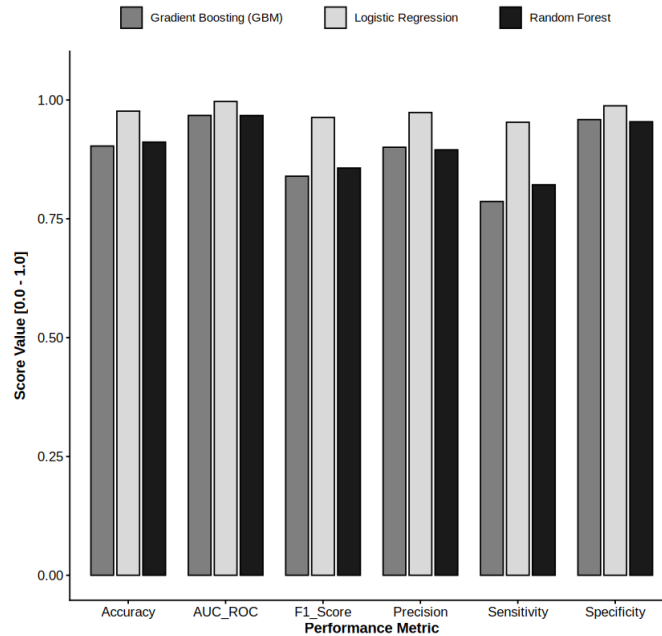


Fig. 12. Comparative validation performance across models



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9.3 ROC Analysis

The ROC curves in Fig. 13 plot sensitivity against specificity for all thresholds. All three models lie far above the diagonal of a random classifier. Random Forest (AUC = 0.967) and GBM (AUC = 0.968) are almost identical, while Logistic Regression reaches 0.997. An AUC above 0.96 means that a randomly chosen at-risk student receives a higher risk score than a randomly chosen normal student more than 96% of the time.

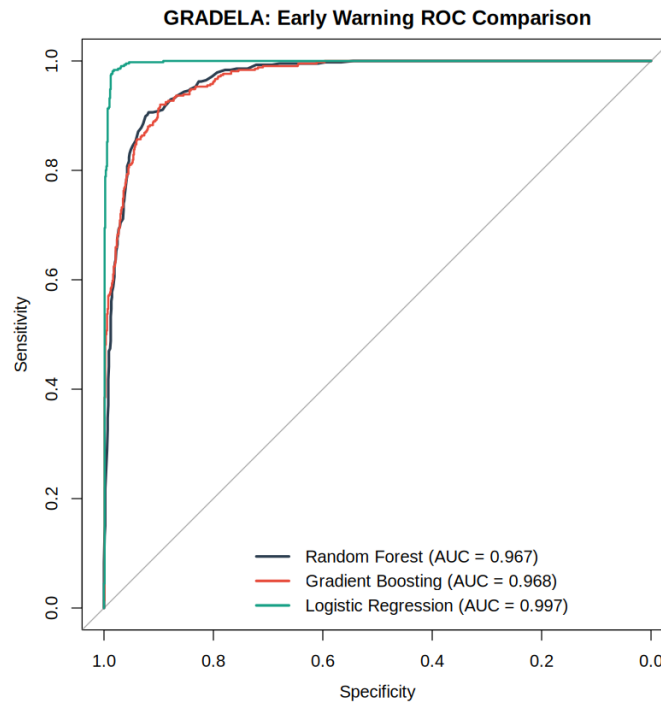


Fig. 13. ROC curve comparison of the three classifiers

9.4 Cost-Sensitive Threshold Calibration

By default a classifier flags a student when $P(\text{High_Risk}) \geq 0.50$. In academic intervention, however, the costs are asymmetric: a False Negative means a struggling student receives no help, while a False Positive simply means a student is offered extra tutoring. `R/04_threshold_tuning.R` evaluates sensitivity across the cut-offs $\tau \in \{0.50, 0.44, 0.38, 0.30\}$ and selects $\tau^* = 0.38$ as the point that raises recall without flooding advisors with false alarms.



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Effect of Cost-Sensitive Threshold Calibration (Random Forest)

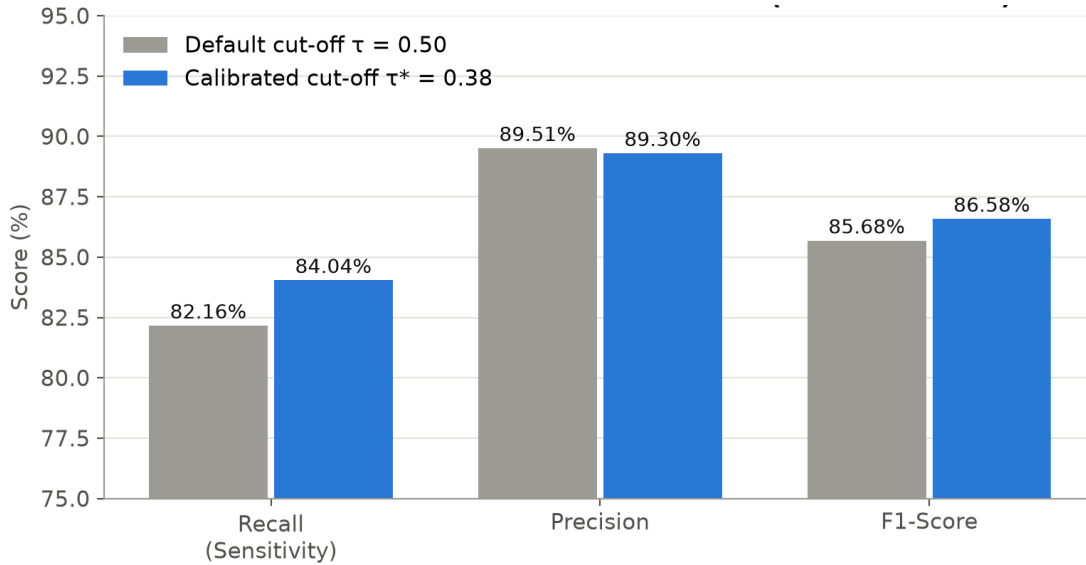


Fig. 14. Effect of lowering the decision threshold from 0.50 to 0.38

At $\tau^* = 0.38$, recall rises from 82.16% to **84.04%** and the F1-score improves from 0.8568 to **0.8658**, while precision falls only marginally (89.51% $\rightarrow$ 89.30%). The ROC-AUC is unchanged because it is threshold-independent. Lowering τ further to 0.30 would capture more students but at the cost of a sharp rise in false positives, which would dilute advisor capacity.



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9.5 Continuous Score Forecasting

In addition to the binary flag, the Random Forest regressor predicts each student's exact exam score. On the test partition it achieved $R^2 = 0.632$, **RMSE = 2.40 marks** and **MAE = 1.08 marks**. Fig. 15 shows that predictions lie close to the 45° line for the bulk of students; the few large residuals correspond to the unusual scores above 85.

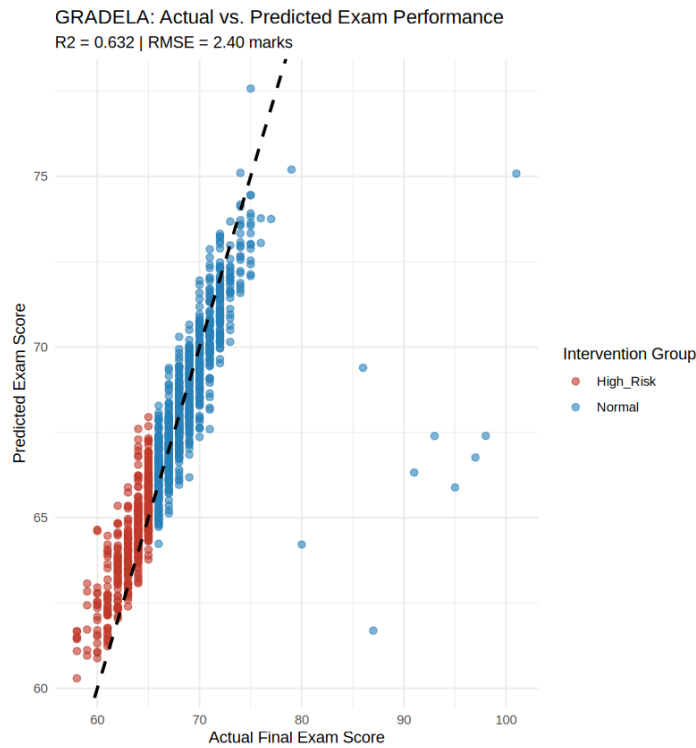


Fig. 15. Actual versus predicted exam scores (Random Forest regressor)

Table 11. Score regression metrics

Metric	Value	Meaning
R^2	0.6322	63.2% of exam-score variance explained
RMSE	2.40 marks	Typical error, penalising large misses
MAE	1.08 marks	Average absolute deviation on a 100-point scale



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10. FEATURE IMPORTANCE AND INTERPRETABILITY

A BI system is only useful if decision-makers trust and understand it. GRADELA therefore provides two layers of interpretability: global feature importance from the Random Forest and local, rule-based explanations from a decision tree.

10.1 Random Forest Feature Importance

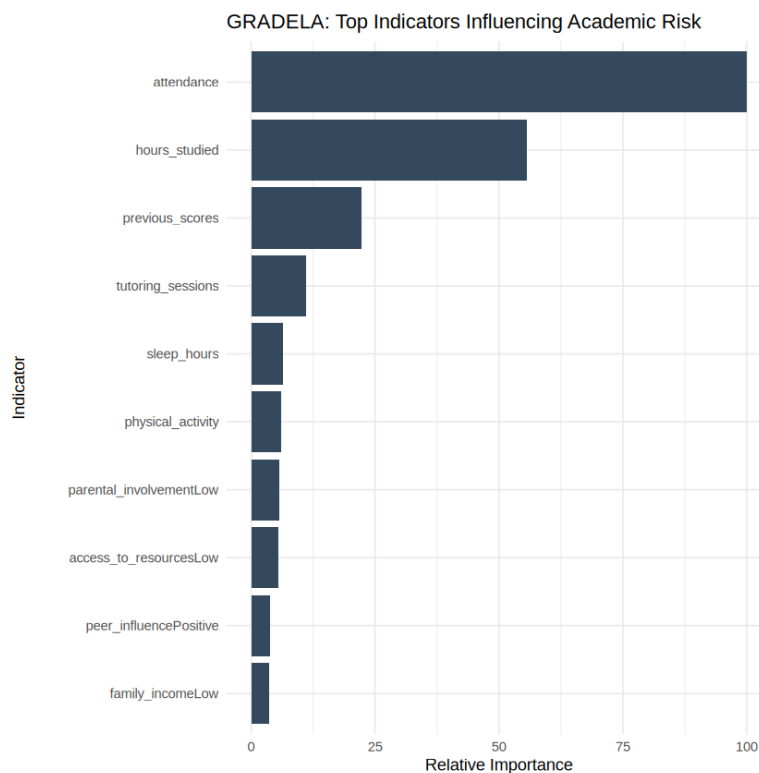


Fig. 16. Top indicators influencing academic risk (Random Forest varImp)

Fig. 16 ranks the predictors by relative importance. **Attendance** is by far the most influential indicator, followed by **Hours_Studied** and **Previous_Scores**. Tutoring sessions, sleep hours and physical activity form a secondary group, and categorical context such as low parental involvement or low access to resources contributes smaller increments. Static demographics such as gender and school type barely register.

The practical implication is significant: **the strongest drivers of risk are behaviours that an institution can influence** – attendance and study habits – rather than fixed characteristics that it cannot change.



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10.2 Classroom Intervention Rules

The CART tree in Fig. 17 converts the model logic into simple rules. The root split is attendance < 74%: 65% of students below this level are at risk. Among them, those studying fewer than 23 hours per week form the highest-risk leaf (81% at risk, 22% of all students). Students with attendance $\geq 74\%$ and at least 17 study hours per week – almost half of the cohort (48%) – are almost always safe, with only 7% at risk.

GRADELA: Early Intervention Decision Tree

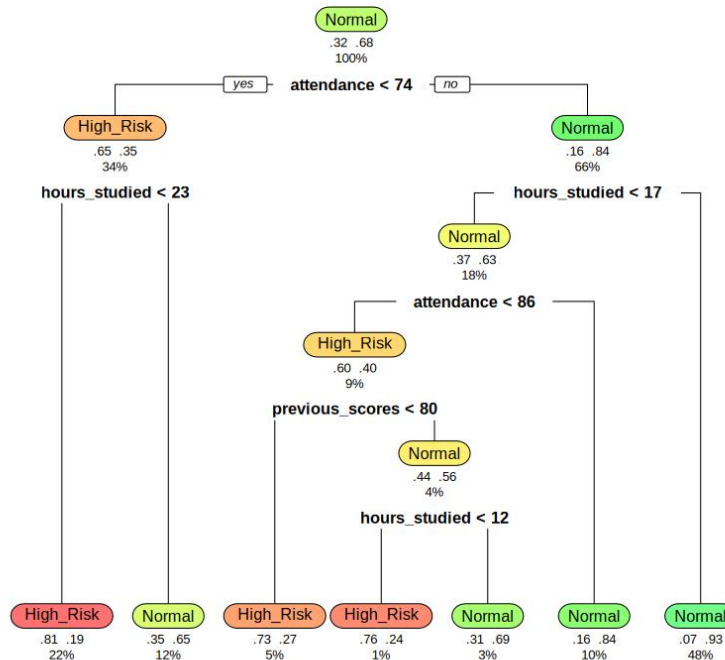


Fig. 17. Early-intervention decision tree (rpart, cp = 0.015)

Table 12. Human-readable intervention rules derived from the decision tree

Rule	Condition	Outcome
R1	Attendance < 74% AND study hours < 23	High_Risk (81% probability) – immediate outreach
R2	Attendance < 74% AND study hours ≥ 23	Normal (65%) – monitor attendance
R3	Attendance $\geq 74\%$ AND study hours < 17 AND attendance < 86% AND previous score < 80	High_Risk (73%) – study-skills support
R4	Attendance $\geq 74\%$ AND study hours ≥ 17	Normal (93%) – standard monitoring



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11. RISK STRATIFICATION AND INTERVENTION ANALYSIS

The final stage of GRADELA (R/05_alert_dispatch.R) turns probabilities into decisions. Every student in the scored cohort receives a risk probability, a risk tier, a primary risk driver and a recommended intervention.

11.1 Advising Protocol

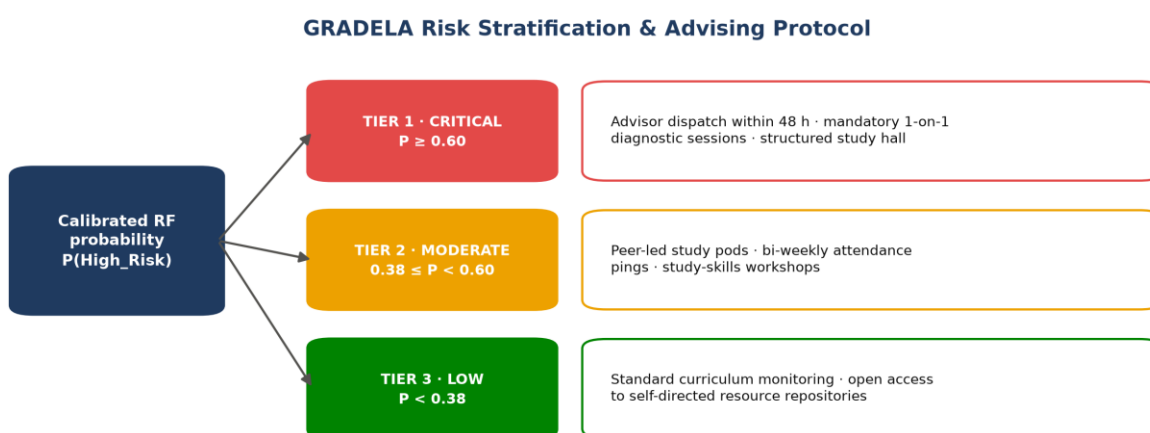


Fig. 18. Three-tier risk stratification and advising protocol

Table 13. Risk tiers and operational advising protocols

Tier	Probability Range	Classification	Advising Protocol
Tier 1	$P \geq 0.60$	High / Critical Risk	Advisor dispatch within 48 h; mandatory 1-on-1 diagnostic sessions; structured study hall
Tier 2	$0.38 \leq P < 0.60$	Moderate Risk	Peer-led study pods; bi-weekly attendance pings; study-skills workshops
Tier 3	$P < 0.38$	Low Risk / On-Track	Standard curriculum monitoring; self-directed resource access

11.2 Tier Distribution

Applying the protocol to the scored test cohort of 1,322 students produced **282 Tier 1** students (21.3%), **192 Tier 2** students (14.5%) and **848 Tier 3** students (64.1%). In total, **474 students (35.9%)** are flagged for some form of intervention and exported to gradela_intervention_roster.csv.



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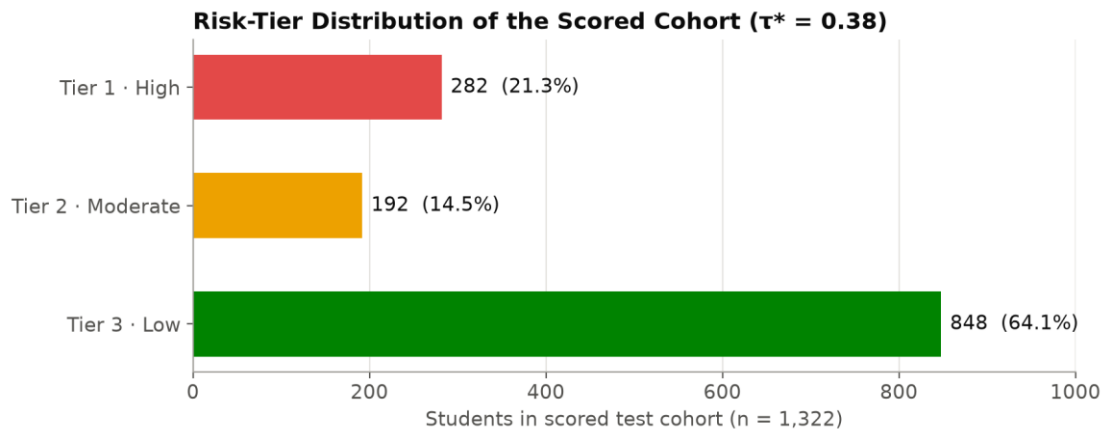


Fig. 19. Risk-tier distribution of the scored cohort

11.3 Primary Risk Drivers

For each student, the dispatch script identifies the dominant deficit. Fig. 20 shows that **Severe Attendance Deficit** is the most common driver overall (368 students) and dominates Tier 1 (219 of 282 critical students). **Insufficient Study Volume** is the second most common driver and is spread across all tiers. Low-risk students are mostly labelled "Multivariate Risk Factor", meaning no single deficit stands out.

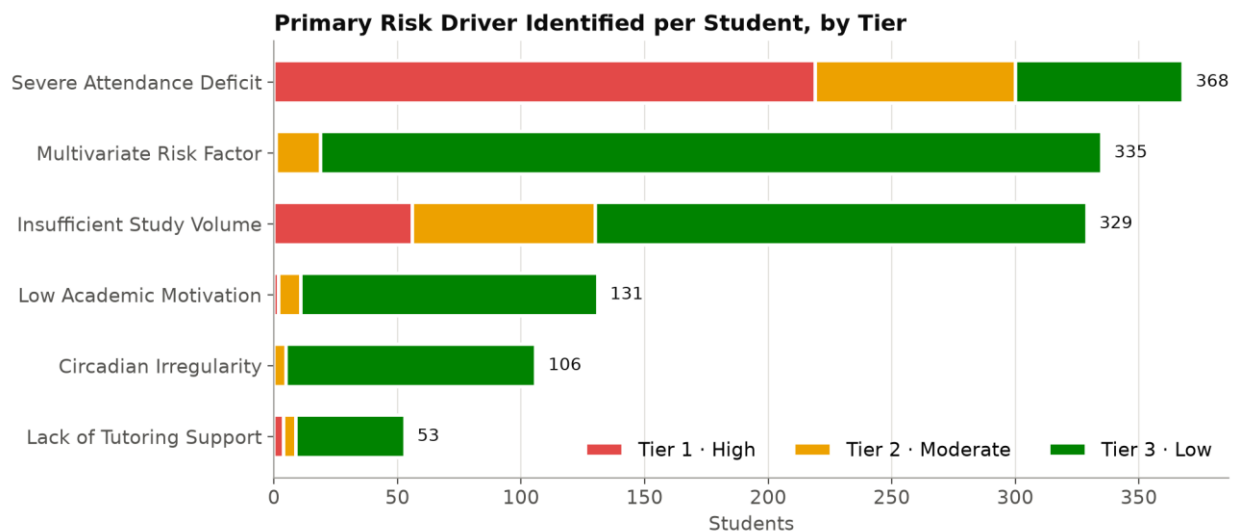


Fig. 20. Primary risk driver identified per student, by tier



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11.4 Tier Behaviour Profile

Fig. 21 confirms that the tiers are behaviourally meaningful. Average attendance falls from 85.6% in Tier 3 to 73.9% in Tier 2 and 67.6% in Tier 1, while weekly study hours fall from 21.5 to 18.6 and 16.0. Previous scores and tutoring follow the same gradient, although the differences are smaller.

Average Behavioural Profile of Each Risk Tier

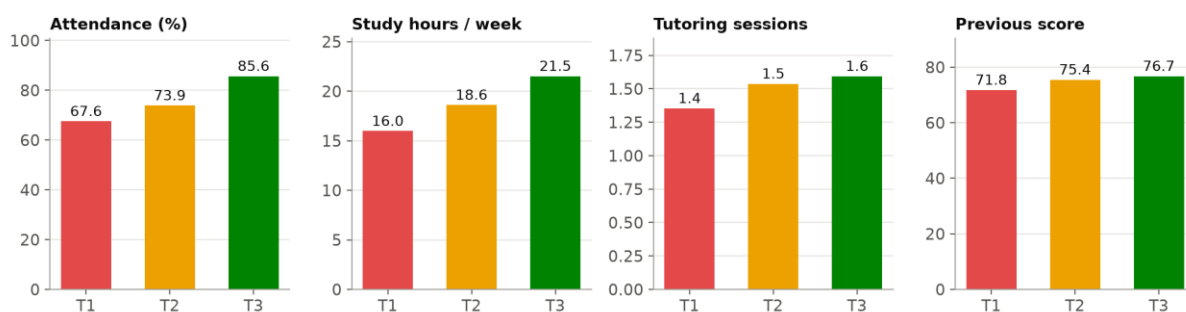


Fig. 21. Average behavioural profile of each risk tier

11.5 Intervention Roster Output

The roster is sorted by descending risk probability so that advisors can work from the top. A sample of the output is shown below.

Table 14. Sample of the generated student risk alert roster

Student ID	Risk Prob.	Risk Tier	Primary Driver	Recommended Intervention
ST-1213	0.98	High (Tier 1)	Severe Attendance Deficit	Immediate 1-on-1 advising within 48 h; mandatory study hall
ST-0755	0.96	High (Tier 1)	Severe Attendance Deficit	Immediate 1-on-1 advising within 48 h; mandatory study hall
ST-0003	0.62	High (Tier 1)	Severe Attendance Deficit	Immediate 1-on-1 advising within 48 h; mandatory study hall
ST-0002	0.40	Moderate (Tier 2)	Insufficient Study Volume	Peer collaborative learning pod; bi-weekly attendance ping
ST-0001	0.12	Low (Tier 3)	Insufficient Study Volume	Standard monitoring; self-directed portal access



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12. DASHBOARD AND WEB APPLICATION

To make GRADELA usable by non-technical staff, the analytics are delivered through two interfaces: an R Shiny LMS early-warning dashboard (R/shiny_app.R) for institutional use with the R models, and the **GRADELA AI** web application, publicly hosted at <https://gradela.vercel.app/>. The web application is a companion decision-support interface that runs its machine-learning models entirely in the browser (client-side, offline-capable) on the UCI Student Performance benchmark (Cortez & Silva, 2008), demonstrating how the same early-warning concept can be packaged for end users.

12.1 Live Predictor and Risk Assessment

The landing page (Fig. 22) lets an advisor enter or adjust a student's profile using sliders and toggles, or load preset personas (high achiever, average, at-risk). Two prediction horizons are offered: **Early Warning Mode** (no periodic grades) and **Mid-Term Mode** (with G1 and G2 grades).

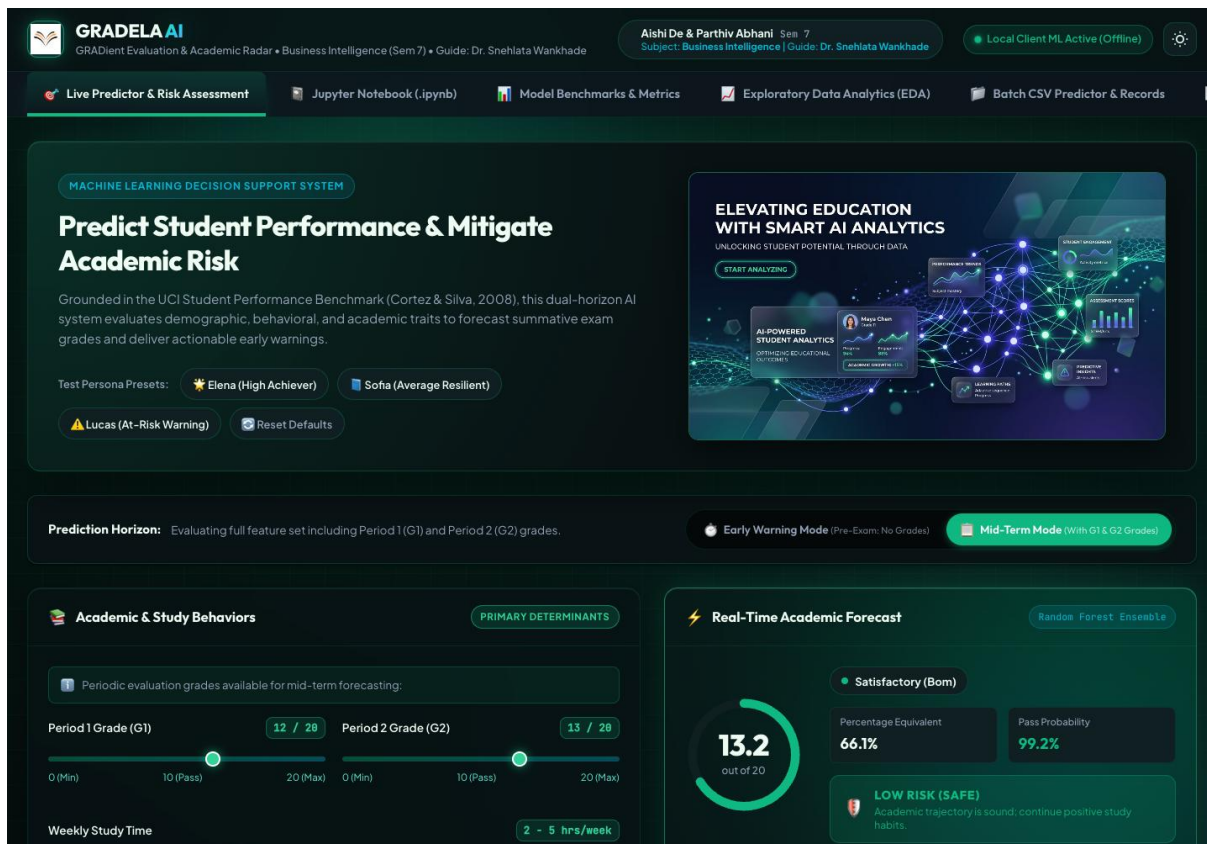


Fig. 22. GRADELA AI web application – live predictor landing page



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Fig. 23 shows the forecast for an at-risk persona in Early Warning mode. The engine predicts a grade of 5.8/20 (29%) with a pass probability of 2.7%, raises a **CRITICAL RISK** banner, decomposes the result into four predictive pillars, lists the identified risk factors and generates an actionable pedagogical intervention plan.

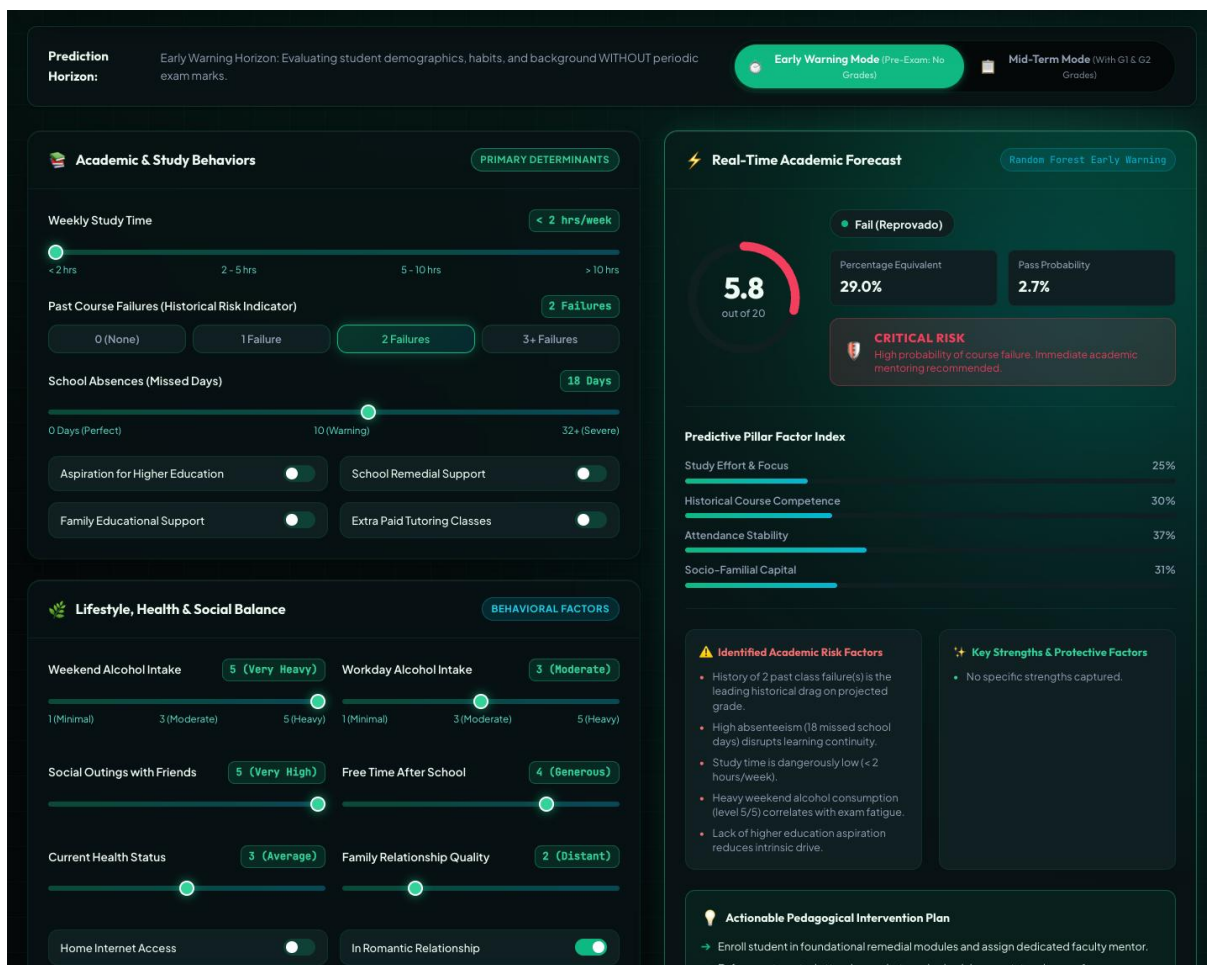


Fig. 23. Real-time academic forecast for an at-risk student



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12.2 Model Benchmarks and Exploratory Analytics

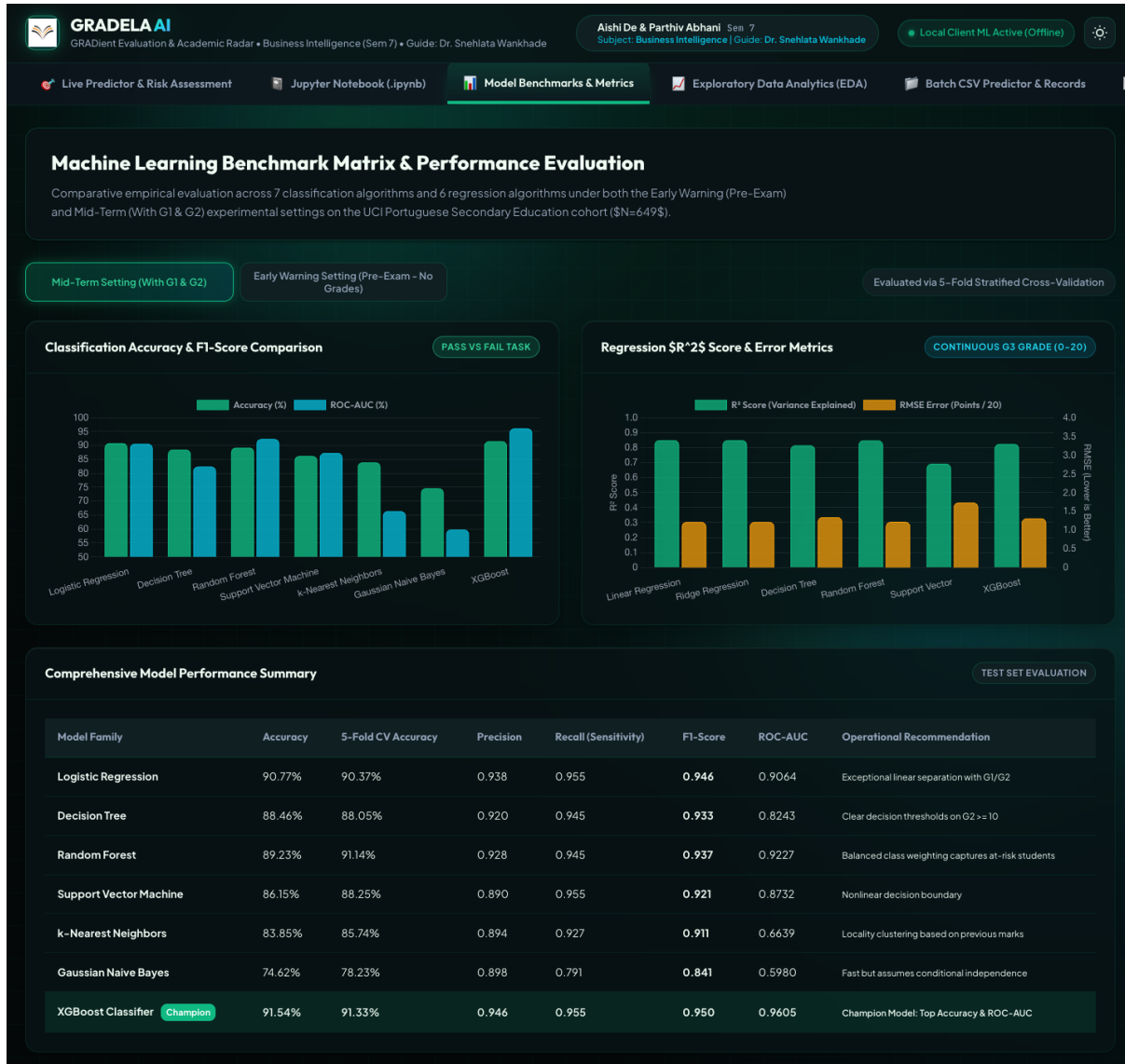


Fig. 24. Model benchmark matrix and performance evaluation tab

The benchmark tab (Fig. 24) compares seven classification and six regression algorithms under both prediction horizons, with an accuracy/ROC-AUC chart, a regression error chart and a detailed metrics table. The EDA tab (Fig. 25) visualises the grade distribution, study-time effects, the impact of past failures and other empirical trends.



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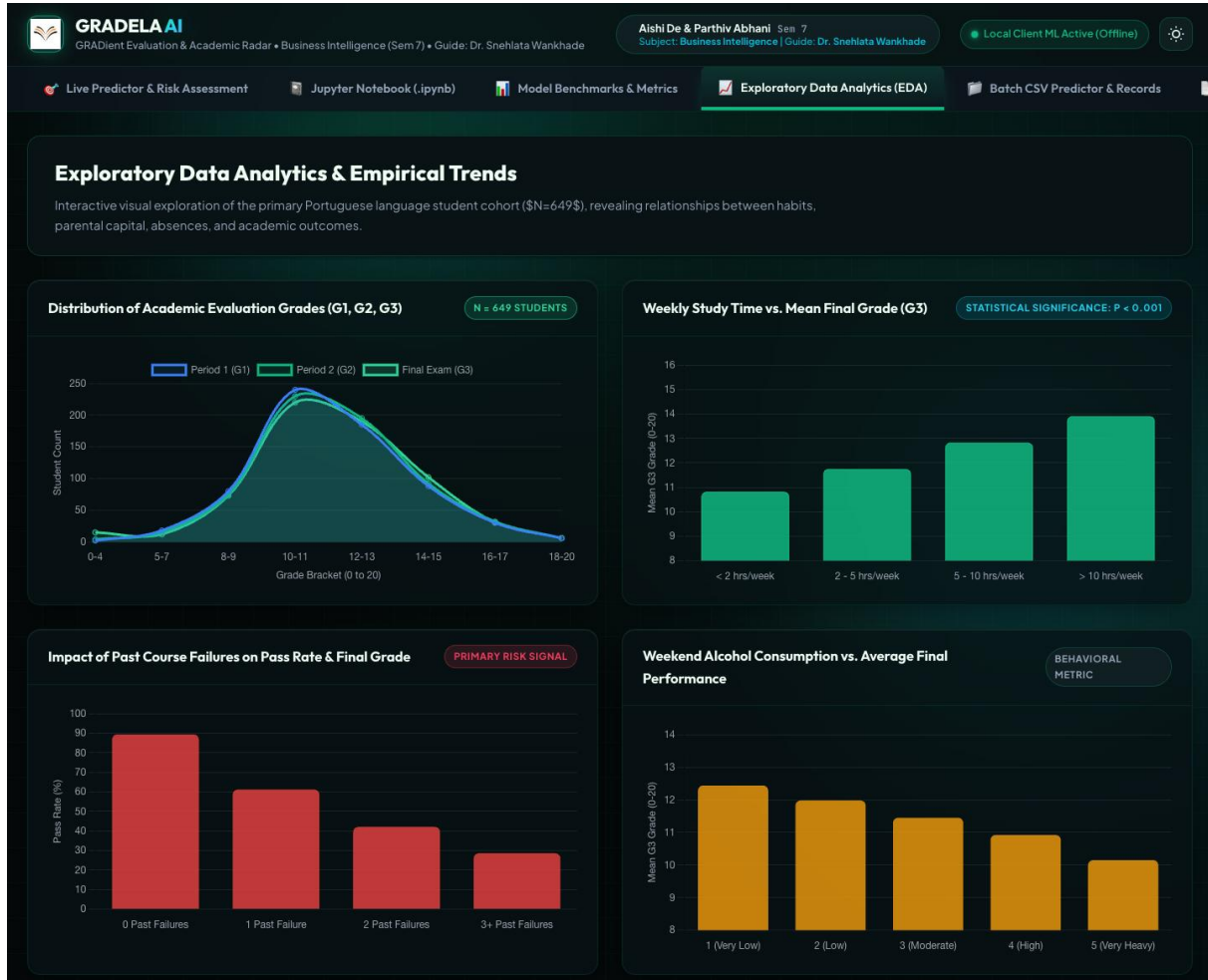


Fig. 25. Exploratory data analytics tab

12.3 Batch Inference, Research and Notebook Views

The batch tab (Fig. 26) allows an institution to upload a CSV of students and score the whole cohort at once, producing a downloadable student-records ledger. The research tab (Fig. 27) documents the hypotheses and literature, and the notebook tab (Fig. 28) embeds the complete GRADELA R notebook so that every result is traceable to code.



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**GRADELA AI**
GRADient Evaluation & Academic Radar • Business Intelligence (Sem 7) • Guide: Dr. Snehlata Wankhade

Aishi De & Parthiv Abhani Sem: 7
Subject: Business Intelligence | Guide: Dr. Snehlata Wankhade

Local Client ML Active (Offline)

Live Predictor & Risk Assessment

Jupyter Notebook (.ipynb)

Model Benchmarks & Metrics

Exploratory Data Analytics (EDA)

Batch CSV Predictor & Records

Batch CSV Inference Engine & Institutional Student Records

Upload student records in CSV format or examine institutional sample cohorts from the UCI repository. The system executes automated inference across every record, flagging at-risk individuals.

**Drag & Drop Student CSV File** or
[browse from computer](#)
Accepts semicolon (,) or comma (,) separated files matching UCI format.

 **Load UCI Benchmark Sample (40 Students)**

 **Export Scored Predictions CSV**

Student Records & Inference Ledger Showing 0 records

Filter by school, sex, risk level... All Risk Levels

ID	School	Sex / Age	Study Time	Failures	Absences	G1/ G2	Actual G3	Predicted G3	Pass Probability	Risk Status
----	--------	-----------	------------	----------	----------	--------	-----------	--------------	------------------	-------------

Click "Load UCI Benchmark Sample" or upload a CSV file to evaluate student records.

Fig. 26. Batch CSV inference engine and student records ledger

**GRADELA AI**
GRADient Evaluation & Academic Radar • Business Intelligence (Sem 7) • Guide: Dr. Snehlata Wankhade

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Live Predictor & Risk Assessment

Jupyter Notebook (.ipynb)

Model Benchmarks & Metrics

Exploratory Data Analytics (EDA)

Batch CSV Predictor & Records

Research Hypothesis, Theoretical Foundations & Literature Review

Detailed documentation of the research hypothesis formulated in Continuous Assessment I (CA I), statistical validation methodology, and alignment with foundational Educational Data Mining literature.

Research Hypotheses & Statistical Confirmation

EMPIRICALLY VALIDATED

Alternative Hypothesis (SH_15) [Accepted]:
"Students' demographic, academic, and behavioral attributes have a statistically significant relationship with their final academic performance (G3). Machine learning algorithms can accurately predict student performance and assist educational institutions in identifying students who require academic support."

Null Hypothesis (SH_05) [Rejected]:
"Students' demographic, academic, and behavioral attributes have no statistically significant relationship with their final academic performance, and machine learning models cannot reliably predict outcomes."

Statistical Hypothesis Testing Results Summary

Statistical Test	Attributes Tested	Test Statistic	p-value	Conclusion
Welch's Two-Sample t-test	Desire for Higher Education (Higher: yes vs no)	$t = 6.2714$	$p = 8.12e-10$	Reject H0 ($p < 0.001$)
Welch's Two-Sample t-test	Weekly Study Time (≥ 3 hrs vs < 3 hrs)	$t = 5.1839$	$p = 2.91e-07$	Reject H0 ($p < 0.001$)
One-Way ANOVA	Mother's Education (Medu tiers 0 to 4)	$F = 10.7643$	$p = 1.89e-08$	Reject H0 ($p < 0.001$)
One-Way ANOVA	Number of Past Class Failures (Failures)	$F = 48.3512$	$p = 1.44e-28$	Reject H0 ($p < 0.001$)
Chi-Square (χ^2) Test	Internet Access at Home vs Pass/Fail	$\chi^2 = 11.238$ (df=1)	$p = 0.0008$	Reject H0 ($p < 0.01$)
Chi-Square (χ^2) Test	Past Failures vs Pass/Fail Outcome	$\chi^2 = 72.841$ (df=3)	$p = 1.04e-15$	Reject H0 ($p < 0.001$)

Related Work & Literature Review Matrix (From CA I)

20 BENCHMARK STUDIES

Synthesized literature review benchmarking our proposed solution against landmark Educational Data Mining publications:

Fig. 27. Research hypotheses, statistical tests and literature review tab

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Local Client ML Active (Offline)

Live Predictor & Risk AssessmentJupyter Notebook (.ipynb)Model Benchmarks & MetricsExploratory Data Analytics (EDA)Batch CSV Predictor & Records

COMPLETE EXECUTED ARTIFACTR Kernel (ip)

GRADELA Pipeline Executed

GRADELA R Jupyter Notebook Solution

Full interactive preview of the GRADELA R programming notebook: containing complete 6.607-student cohort ingestion, exploratory correlation matrix, ensemble benchmarking (Random Forest, GBM, GLM), ROC-AUC evaluation, decision boundary tuning ($t^* = 0.38$), continuous score regression, and classroom intervention rules.

Download .ipynbDownload .R ScriptOpen in New TabRefreshFull Screen

Subject: Business Intelligence (Sem 7)
Guide: Dr. Snehlata Wankhade
Investigators: Aishi De (23070521008) & Parthiv Abhani (23070521106)
Status: Pre-Rendered with all Outputs & Plots

GRADELA_GRADient_Evaluation_& Learned_Academic_Performance_Radar_(Aishi_De_& Parthiv_Abhani).ipynb (R Kernel)Executed R Notebook (.ipynb)Raw R Script (.R)

PROJECT: GRADELA - GRADient Evaluation & Learned Academic Performance Radar

Aishi De & Parthiv Abhani
Sem 7 (23070521008 & 23070521106)
End-to-End Analysis, Predictive Modeling, and Evaluation Pipeline
RUNTIME: R

```
In [2]: required_packages <- c(
  "tidyverse", "caret", "randomForest", "gbm",
  "pROC", "corrplot", "rpart", "rpart.plot", "e1071"
)

installed <- installed.packages()[, "Package"]
```

Fig. 28. Embedded GRADELA R notebook viewer



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13. FIVE MAJOR FINDINGS

Finding 1: Attendance is the Strongest Early-Warning Signal

Attendance has the highest correlation with exam score ($r = 0.58$), the highest Random Forest importance and forms the root split of the decision tree. At-risk students attend 70.8% of classes on average versus 84.4% for others, and Severe Attendance Deficit is the primary driver for 219 of the 282 critical-tier students.

Finding 2: Risk is Predominantly Behavioural, not Demographic

Attendance, study hours and previous scores dominate the importance ranking, while gender and school type show almost no difference in risk rate ($\approx 32\%$ in every group). Most of the risk is therefore driven by modifiable habits.

Finding 3: At-Risk Students Can be Identified with High Reliability

The Random Forest achieved 91.14% accuracy, 95.42% specificity and an ROC-AUC of 0.967 on unseen students, beating the 67.75% no-information baseline by more than 23 percentage points.

Finding 4: Threshold Calibration Improves Coverage at Negligible Cost

Lowering the decision threshold to $\tau^* = 0.38$ increased recall from 82.16% to 84.04% and improved F1 to 0.8658, while precision dropped by only 0.21 percentage points.

Finding 5: Home Environment Amplifies Risk

Low parental involvement (42.1%), low access to resources (42.0%), learning disabilities (43.0%) and lack of internet access (40.3%) all raise the at-risk rate well above the 32.3% cohort average.



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14. THREE MAJOR PROBLEMS

Problem 1: A Large Share of the Cohort Requires Support

About one-third of students (32.25%) fall at or below the risk cut-off, and 474 of 1,322 scored students (35.9%) are flagged for intervention. This volume may exceed the capacity of advisors if every case is handled one-on-one.

Problem 2: Some At-Risk Students are Still Missed

Even after calibration, roughly 16% of at-risk students are not flagged. These students tend to have moderate attendance and study hours, so their risk is driven by a combination of weaker signals that the model finds harder to separate.

Problem 3: Model Selection Trade-off

Logistic Regression outperformed the ensembles on this dataset (AUC 0.997 vs 0.967), indicating a largely linear structure. Deploying the Random Forest favours robustness and explainability for future cohorts but leaves some predictive accuracy unused on the current data.



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15. FOUR RECOMMENDATIONS

Recommendation 1: Institutionalise Attendance Monitoring

Because attendance is the dominant driver, institutions should track it weekly and trigger automatic outreach as soon as a student drops below 74% – the threshold identified by the decision tree.

Recommendation 2: Use Tiered, Capacity-Aware Interventions

Reserve one-on-one advising for the 282 Tier 1 students and serve the 192 Tier 2 students through peer study pods and workshops. This concentrates scarce advisor time where the probability of failure is highest.

Recommendation 3: Target Study-Skills and Resource Gaps

Provide structured study plans, library and internet access, and tutoring slots for students with low study hours, low resource access or learning disabilities, since these groups show elevated risk.

Recommendation 4: Adopt a Champion–Challenger Model Strategy

Run Logistic Regression alongside the Random Forest each term, compare their recall at τ^* , and promote whichever performs better on the new cohort. Re-calibrate τ^* every semester using actual outcomes.



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16. CONCLUSION

GRADELA successfully demonstrates how Business Intelligence and machine learning can move academic support from a reactive, post-examination activity to a proactive, early-semester process. Starting from 6,607 raw student records, the project built a reproducible R pipeline that cleans and enriches the data, defines a data-driven risk target, benchmarks multiple classifiers, calibrates the decision threshold for asymmetric costs, forecasts exact scores and converts every prediction into a tiered intervention.

The production Random Forest identifies at-risk students with 91.14% accuracy and an ROC-AUC of 0.967, and the calibrated threshold $\tau^* = 0.38$ captures 84.04% of at-risk students. Interpretability analysis shows that attendance and study habits – both modifiable – are the principal drivers of risk, which gives institutions a clear and actionable focus. The results are delivered through an intervention roster, an R Shiny dashboard and a hosted web application, completing the BI value chain from data to decision.

Overall, the project shows that predictive analytics does not replace teachers and advisors; it gives them earlier, sharper information so that their limited time is spent on the students who need it most.



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17. FUTURE SCOPE

The current system provides a strong foundation that can be extended in several directions (Fig. 29):

- **LMS integration:** expose the R models through a Plumber API or Shiny Server so that risk scores refresh automatically from the institution's learning management system.
- **Longitudinal re-scoring:** re-run predictions after each assessment so that risk is tracked as a trajectory rather than a single snapshot.
- **Explainable AI:** add SHAP values to give a personalised explanation for every flagged student.
- **Fairness auditing:** monitor model performance across gender, income and school-type groups to ensure equitable support.
- **Outcome feedback loop:** record which interventions were delivered and whether the student recovered, enabling the institution to measure the causal impact of each support programme.
- **Mobile notifications:** push alerts to advisors and students when risk tier changes.

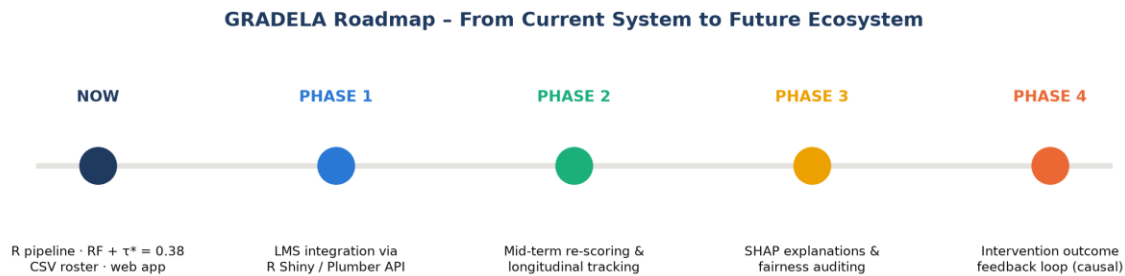


Fig. 29. Roadmap from the current GRADELA system to a future ecosystem



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18. TOOLS AND TECHNOLOGIES USED

Table 15. Tools and technologies used

Tool / Technology	Purpose
R (≥ 4.2)	Core language for data preparation, modelling and evaluation
tidyverse (dplyr, ggplot2)	Data manipulation and visualisation
caret	Stratified partitioning, cross-validation, model training
randomForest, gbm, rpart	Random Forest, Gradient Boosting and CART algorithms
pROC, ROCR	ROC curves and AUC computation
corrplot, rpart.plot	Correlation heatmap and decision-tree visualisation
yaml	Pipeline configuration (config.yml)
Shiny, shinydashboard, DT	Interactive LMS early-warning dashboard
Jupyter Notebook (R kernel)	Documented, reproducible analysis notebook
HTML, CSS, JavaScript	GRADELA AI web application front end
Vercel	Hosting of the web application (gradela.vercel.app)
Git & GitHub	Version control and collaboration
Microsoft PowerPoint	Project presentation

R was the primary analytical platform. The **caret** package provided a unified interface for training and validating all classifiers, while **randomForest**, **gbm** and **rpart** supplied the algorithms. **pROC** was used for ROC analysis, and **Shiny** together with a static **HTML/JavaScript** front end hosted on **Vercel** delivered the results to end users.



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19. CONTRIBUTION OF TEAM MEMBERS

Table 16. Contribution of team members

Member	Technical Focus	Deliverables
Aishi De PRN: 23070521008 Sem 7, Sec A	Machine learning architecture, SEI formulation and behavioural modelling	SEI formulation and model benchmarking (CART, GLM); feature synthesis (02_feature_engineering.R) and model training suite (03_model_training.R); presentation content and methodology write-up; technical walkthrough video
Parthiv Abhani PRN: 23070521106 Sem 7, Sec B	Data engineering, pipeline orchestration and threshold calibration	Stratified cross-validation design and cost-sensitive threshold optimisation ($\tau^* = 0.38$); preprocessing (01_data_preprocessing.R) and alert dispatch (05_alert_dispatch.R); architecture diagrams and visual formatting; live pipeline and dashboard demonstration video

Faculty Guide: Dr. Snehlata Wankhade | **Course:** Business Intelligence (CA-3 Mini Project)

Hosted Web Application: <https://gradela.vercel.app/>

GitHub Repository: <https://github.com/aishide/Gradela>



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APPENDIX A - R PIPELINE CODE

The following listings show the core of each pipeline stage. The complete source is available in the project repository.

A.1 Preprocessing and risk formulation (01_data_preprocessing.R)

```
names(df_raw) <- tolower(gsub("[.]", "_", names(df_raw)))
df_clean <- df_raw %>%
  mutate(across(where(is.numeric), ~ ifelse(is.na(.), median(., na.rm = TRUE), .))) %>%
  mutate(across(where(is.character), ~ ifelse(is.na(.) | trimws(.) == "", "Unknown", trimws(.)))) %>%
  mutate(across(where(is.character), as.factor))

risk_threshold <- as.numeric(quantile(df_clean$exam_score, 0.25)) # 65
df_model <- df_clean %>% mutate(risk_status = factor(
  ifelse(exam_score <= risk_threshold, "High_Risk", "Normal"), levels = c("High_Risk", "Normal")))

set.seed(42)
train_idx <- createDataPartition(df_model$risk_status, p = 0.80, list = FALSE)
```

A.2 Feature synthesis (02_feature_engineering.R)

```
eps <- 1e-4
df_fe <- df_clean %>% mutate(
  sleep_variance = abs(sleep_hours - 7.0) + 0.80,
  study_efficiency_index = round((hours_studied * previous_scores) / (sleep_variance + eps), 2),
  attendance_tutoring_synergy = round(attendance * (tutoring_sessions + 1), 2),
  academic_engagement_ratio = round((hours_studied + tutoring_sessions * 2) / (sleep_hours + 1), 2))
```

A.3 Model training (03_model_training.R)

```
cv_ctrl <- trainControl(method = "cv", number = 5, classProbs = TRUE,
  summaryFunction = twoClassSummary, savePredictions = "final")
fit_glm <- train(x = train_features, y = train_risk, method = "glm", family = "binomial",
  metric = "ROC", trControl = cv_ctrl)
fit_rf <- train(x = train_features, y = train_risk, method = "rf", ntree = 100,
  metric = "ROC", trControl = cv_ctrl)
fit_gbm <- train(x = train_features, y = train_risk, method = "gbm", metric = "ROC",
  trControl = cv_ctrl, verbose = FALSE, tuneLength = 2)
fit_cart <- rpart(risk_status ~ ., data = cart_data, method = "class", control = rpart.control(cp = 0.015))
fit_reg_rf <- randomForest(exam_score ~ ., data = reg_data, ntree = 120, importance = TRUE)
```

A.4 Threshold tuning (04_threshold_tuning.R)

```
prob_high_risk <- predict(fit_rf, test_features, type = "prob")$High_Risk
for (tau in c(0.50, 0.44, 0.38, 0.30)) {
  pred_tau <- factor(ifelse(prob_high_risk >= tau, "High_Risk", "Normal"),
    levels = c("High_Risk", "Normal"))
  cm_tau <- confusionMatrix(pred_tau, test_labels, positive = "High_Risk")
  roc_obj <- roc(test_labels, prob_high_risk, levels = c("Normal", "High_Risk"))
}
```

A.5 Risk tier assignment (05_alert_dispatch.R)

```
risk_tier = case_when(
  risk_probability >= 0.60 ~ "High Risk (Tier 1)",
  risk_probability >= 0.38 ~ "Moderate Risk (Tier 2)",
  TRUE ~ "Low Risk (Tier 3)")
```



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APPENDIX B - DATA DICTIONARY

Table 17. Processed cohort data dictionary

Field (snake_case)	R Type	Analytical Role
hours_studied	integer	Behavioural predictor – study volume
attendance	integer	Behavioural predictor – class presence
parental_involvement	factor	Home-support context
access_to_resources	factor	Resource context
extracurricular_activities	factor	Lifestyle context
sleep_hours	integer	Lifestyle predictor; input to SEI
previous_scores	integer	Academic history; input to SEI
motivation_level	factor	Personal context
internet_access	factor	Resource context
tutoring_sessions	integer	Support-service usage
family_income	factor	Socio-economic context
teacher_quality	factor	School context (incl. Unknown level)
school_type	factor	School context
peer_influence	factor	Social context
physical_activity	integer	Lifestyle predictor
learning_disabilities	factor	Personal context
parental_education_level	factor	Socio-economic context (incl. Unknown level)
distance_from_home	factor	Geographic context (incl. Unknown level)
gender	factor	Demographic attribute
exam_score	integer	Regression target
risk_status	factor	Classification target (High_Risk / Normal)

Table 18. Output fields of student_risk_alerts.csv

Field	Description
student_id	Anonymised student identifier (ST-xxxx)
risk_probability	Random Forest probability of High_Risk
risk_tier	Tier 1 / Tier 2 / Tier 3 assignment
primary_driver	Dominant deficit behind the risk score
recommended_intervention	Advising action mapped from the tier
predicted_status	Predicted class at $\tau^* = 0.38$
predicted_exam_score	Random Forest regressor forecast
study_efficiency_index	Engineered SEI value



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AI REPORT

[Attach Turnitin AI-detection report screenshot here]



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